Suggested Improvements: Real-Time IoT Data Pipeline

Oct 31, 2026 · @Rana Shaaban

The pipeline covers the proposal's scope, and the eight improvements below would make it more accurate, more reliable, and closer to production. They come from the team's literature review and from the limits set in the project scope. Suggestions from the lecturer are recorded in the Feedback and Evaluation document and added here.

1. Current Limits

- Alerts use fixed thresholds, which can raise false alarms or miss slow drift.
- Data is simulated only, with no physical sensors.
- Machine-learning anomaly detection is outside the project scope.
- The batch path loads from batched files rather than directly from Kafka into Snowflake.
- Alert delivery is limited to the alert output and the dashboard history.

2. Proposed Improvements

Effort is the team's rough estimate. The evidence column points to the source in the Literature Review where one exists.

#	Improvement	Why	Effort	Horizon
1	Replace fixed thresholds with dynamic thresholds based on a Smoothed Z-Score	Dynamic thresholds avoid setting temperature limits by experience (Liu et al., 2020); rule-based methods can raise false alarms on noisy data (arXiv 2511.00481)	Medium	Next iteration

#	Improvement	Why	Effort	Horizon
2	Add a statistical baseline, then a machine-learning detector	Threshold rules miss drift that statistical methods catch (Tractian); the survey also lists drift adaptation as an open need (Chatterjee and Ahmed, 2022)	High	Future work
3	Send alert notifications (for example email or a message) in addition to the dashboard	Users learn about a breach faster; team idea, not from a source	Low	Next iteration
4	Load Kafka data into Snowflake directly with the Snowflake connector for Kafka	The vendor states 5 to 10 second end-to-end latency, server-side validation, and error tables for invalid records (Snowflake documentation)	Medium	Next iteration
5	Schedule and orchestrate the batch ETL with a workflow tool (the proposal lists Azure Data Factory)	The proposal aims to introduce orchestration, and a scheduler removes manual ETL runs	Medium	Next iteration
6	Tune Kafka partitions, replication, and acknowledgment settings	These settings affect durability, latency, and consumer lag (Javed and Rajumar, 2026)	Medium	Future work
7	Add automated tests and a GitHub workflow that runs them on every push	Repeatable tests support the testing and documentation criteria; team idea	Medium	Next iteration
8	Connect real sensors in place of the simulator	Moves the project from a simulation to a real deployment; team idea	High	Future work

3. Suggested Order

1. Start with the low-effort items (3 and 7), because they add value quickly.

2. Then improve detection quality (1) and automate the data flow (4 and 5).
3. Leave the high-effort items (2 and 8) and Kafka tuning (6) for later work.